The geometry of Linear Programming

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Outline

The dimension of a polyhedron

Faces, facets and vertices

Basic feasible solutions and vertices

The Fundamental Theorem, geometrically

The dimension of a polyhedron

Introduction: The dimension of a polyhedron

- The feasible region of an LP problem is a polyhedron $F \subseteq \mathbb{R}^n$
- So far we have described it through its **linear constraints**
- In this chapter we describe its **geometric** structure, and we relate it to the algebraic notions of **basis** and **basic feasible solution**
- The first geometric notion is the **dimension**: how many independent directions a polyhedron has

Affine hull and dimension

Definition 1: affine hull and dimension

*Given a polyhedron $F \subseteq \mathbb{R}^n$, its **affine hull** $\mathrm{aff}(F)$ is the smallest affine subspace of $\mathbb{R}^n$ containing F. The **dimension** of F, denoted by $\dim(F)$, is the dimension of $\mathrm{aff}(F)$.*

Equivalently, $\dim(F) = k$ if and only if F contains $k+1$ points $\boldsymbol{x}^{(0)}, \boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(k)}$ such that the k vectors $\boldsymbol{x}^{(1)} - \boldsymbol{x}^{(0)}, \ldots, \boldsymbol{x}^{(k)} - \boldsymbol{x}^{(0)}$ are linearly independent, and it does not contain $k+2$ points with this property.

Full-dimensional polyhedron

Definition 2: full-dimensional polyhedron

*A polyhedron $F \subseteq \mathbb{R}^n$ is called **full-dimensional** if* $\dim(F) = n$. *By convention,* $\dim(\emptyset) = -1$, *and a polyhedron consisting of a single point has dimension* 0.

- The dimension of a polyhedron is not the number of variables: it is the number of independent directions along which one can move inside the polyhedron
- Every equality constraint satisfied by all the points of F, and linearly independent of the other ones, reduces its dimension by one

Standard and canonical forms

- The feasible region of an LP problem in **standard maximization form**,

$$F = \{ \boldsymbol{x} \in \mathbb{R}^n \ : \ \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \ \boldsymbol{x} \geq 0 \},$$

  is never full-dimensional when $F \neq \emptyset$ and $\text{rank}(\boldsymbol{A}) \geq 1$

- All its points satisfy the m equality constraints, and then

$$\dim(F) \ \leq \ n - \text{rank}(\boldsymbol{A})$$

- The feasible region of an LP problem in **canonical maximization form** is instead full-dimensional whenever it contains a point satisfying all the constraints strictly

Let us compute the dimension of two feasible regions:

one in canonical form and one in standard form.

Example: dimension of a polyhedron (1/2)

- The feasible region of the LP problem in canonical maximization form

$$\begin{array}{rrrrrrrr} \max & 5 & x_1 & + & 2 & x_2 & & \\ \text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\ & & x_1 & + & 2 & x_2 & \leq & 12 \\ & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\ & & x_1, & & & x_2 & \geq & 0 \end{array}$$

is **full-dimensional**: $\dim(F) = 2$, since for instance the point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ satisfies all five constraints strictly, and then a small disc around it is contained in F

Example: dimension of a polyhedron (2/2)

- The feasible region of the LP problem in standard maximization form

$$F = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 \ : \ 2x_1 + 3x_2 + x_3 = 4, \quad \boldsymbol{x} \geq 0 \right\}$$

  has dimension $\dim(F) = 2$, and then it is not full-dimensional

- All its points lie on the plane $2x_1 + 3x_2 + x_3 = 4$
- Here $n = 3$, $\text{rank}(\boldsymbol{A}) = 1$ and indeed

$$\dim(F) = 2 = n - \text{rank}(\boldsymbol{A})$$

Summary: the dimension of a polyhedron

Dimension:

- $\dim(F)$ is the dimension of the affine hull of F: the number of independent directions inside F
- $F \subseteq \mathbb{R}^n$ is full-dimensional if $\dim(F) = n$; $\dim(\emptyset) = -1$, a single point has dimension 0

Forms of an LP problem:

- Standard form: $\dim(F) \leq n - \text{rank}(\boldsymbol{A})$, never full-dimensional
- Canonical form: full-dimensional if some point satisfies all the constraints strictly

Faces, facets and vertices

Introduction: Faces, facets and vertices

- A polyhedron has a **boundary** made of pieces of different dimensions: in $\mathbb{R}^2$, edges and corner points
- We describe these pieces through **valid inequalities** and **faces**
- The **facets** are the constraints that cannot be removed, the **vertices** will turn out to be the basic feasible solutions
- We also characterize the vertices as the **extreme points** of the polyhedron

Valid inequality

Definition 3: valid inequality

A linear inequality $\boldsymbol{a}'\boldsymbol{x} \leq \beta$, *with* $\boldsymbol{a} \in \mathbb{R}^n$ *and* $\beta \in \mathbb{R}$, *is* ***valid*** *for a polyhedron* $F \subseteq \mathbb{R}^n$ *if it is satisfied by all its points:*

$$\boldsymbol{a}'\boldsymbol{x} \leq \beta, \qquad \forall \boldsymbol{x} \in F.$$

Face

Definition 4: face

Given a polyhedron $F \subseteq \mathbb{R}^n$ and a valid inequality $\boldsymbol{a}'\boldsymbol{x} \leq \beta$ for F, the set

$$G = \left\{ \boldsymbol{x} \in F \ : \ \boldsymbol{a}'\boldsymbol{x} = \beta \right\}$$

*is called a **face** of F. A face is **proper** if $G \neq \emptyset$ and $G \neq F$.*

Faces of a polyhedron

- A face of a polyhedron is obtained by requiring that one valid inequality is satisfied with equality, and it is itself a polyhedron
- Both F and $\emptyset$ are faces of F:
  - the first one is obtained with the valid inequality $0'\boldsymbol{x} \leq 0$
  - the second one with any valid inequality which is never satisfied with equality

Constraints and faces

- Every linear constraint $\sum_j a_{ij} x_j \leq b_i$ of the LP problem is a valid inequality for its feasible region
- Then it defines a face: the set of the feasible solutions for which the constraint is **binding**
- Together with the Complementary Slackness conditions, this is the link between the geometry of F and the dual problem

Facet and vertex

Definition 5: facet and vertex

Given a polyhedron $F \subseteq \mathbb{R}^n$ and a non-empty face G of F:

1. *G is a **facet** of F if $\dim(G) = \dim(F) - 1$;*
2. *G is a **vertex** of F if $\dim(G) = 0$, that is, if G consists of a single point.*

A facet is always a proper face. A vertex is a proper face, except when F itself consists of a single point: in that case $G = F$ is the only vertex of F.

Why facets and vertices matter

- The facets are the **largest** proper faces
- When the polyhedron has at least one vertex (as the feasible region of an LP problem with non-negative variables), the vertices are the **smallest** non-empty ones
- The facets are important because they are exactly the constraints that cannot be removed: a full-dimensional polyhedron is described by its facet-defining inequalities, and by those only
- The vertices are important because, as we show below, they are exactly the basic feasible solutions

A vertex is defined through a valid inequality.

We now give an equivalent definition,
which uses only the points of the polyhedron.

Extreme point

Definition 6: extreme point

A point $\boldsymbol{x} \in F$ *is an* ***extreme point*** *of the polyhedron* F *if it does not lie strictly between two distinct points of* F*, that is, if there are no* $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in F$ *with* $\boldsymbol{x}^{(1)} \neq \boldsymbol{x}^{(2)}$ *and* $\lambda \in (0, 1)$ *such that*

$$\boldsymbol{x} = \lambda \boldsymbol{x}^{(1)} + (1 - \lambda) \boldsymbol{x}^{(2)}.$$

Vertices and extreme points

Theorem 1: vertices and extreme points

Given a polyhedron $F \subseteq \mathbb{R}^n$ and a point $\boldsymbol{x} \in F$, the following two statements are equivalent:

1. $\{\boldsymbol{x}\}$ *is a vertex of* F*;*
2. $\boldsymbol{x}$ *is an extreme point of* F*.*

- We prove the two implications separately: $(1 \Rightarrow 2)$ uses the valid inequality of the vertex, $(2 \Rightarrow 1)$ uses the constraints binding at $\boldsymbol{x}$

Proof (1/3)

- $(1 \Rightarrow 2)$ Let $\{\boldsymbol{x}\}$ be the face defined by the valid inequality $\boldsymbol{a}'\boldsymbol{y} \leq \beta$, and assume

$$\boldsymbol{x} = \lambda\, \boldsymbol{x}^{(1)} + (1 - \lambda)\, \boldsymbol{x}^{(2)}$$

  with $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in F$ and $\lambda \in (0, 1)$

- Then

$$\beta = \boldsymbol{a}'\boldsymbol{x} = \lambda \underbrace{\boldsymbol{a}'\boldsymbol{x}^{(1)}}_{\leq \beta} + (1 - \lambda) \underbrace{\boldsymbol{a}'\boldsymbol{x}^{(2)}}_{\leq \beta} \leq \beta$$

- The inequality can hold with equality only if $\boldsymbol{a}'\boldsymbol{x}^{(1)} = \boldsymbol{a}'\boldsymbol{x}^{(2)} = \beta$

- Accordingly both points belong to the face $\{\boldsymbol{x}\}$, and then $\boldsymbol{x}^{(1)} = \boldsymbol{x}^{(2)} = \boldsymbol{x}$: the two points are not distinct

Proof (2/3)

- $(2 \Rightarrow 1)$ Let $\boldsymbol{x}$ be an extreme point and let I be the set of the indices of the constraints of F binding at $\boldsymbol{x}$
- If the rows $\{\boldsymbol{a}_i\}_{i \in I}$ did not span $\mathbb{R}^n$, there would be a direction $\boldsymbol{d} \neq 0$ with
$$\boldsymbol{a}_i' \boldsymbol{d} = 0, \qquad \forall i \in I$$
- Then, for $\varepsilon > 0$ small enough, both $\boldsymbol{x} \pm \varepsilon \boldsymbol{d}$ would belong to F
- This contradicts the fact that $\boldsymbol{x}$ is an extreme point, since $\boldsymbol{x}$ is their midpoint

Proof (3/3)

- Accordingly the binding rows span $\mathbb{R}^n$
- The system $\{\boldsymbol{a}_i'\boldsymbol{y} = b_i\}_{i \in I}$ has $\boldsymbol{x}$ as its unique solution
- The valid inequality obtained by summing the binding constraints defines the face $\{\boldsymbol{x}\}$

□

Let us look at the faces, facets and vertices
of the first feasible region of the dimension example.

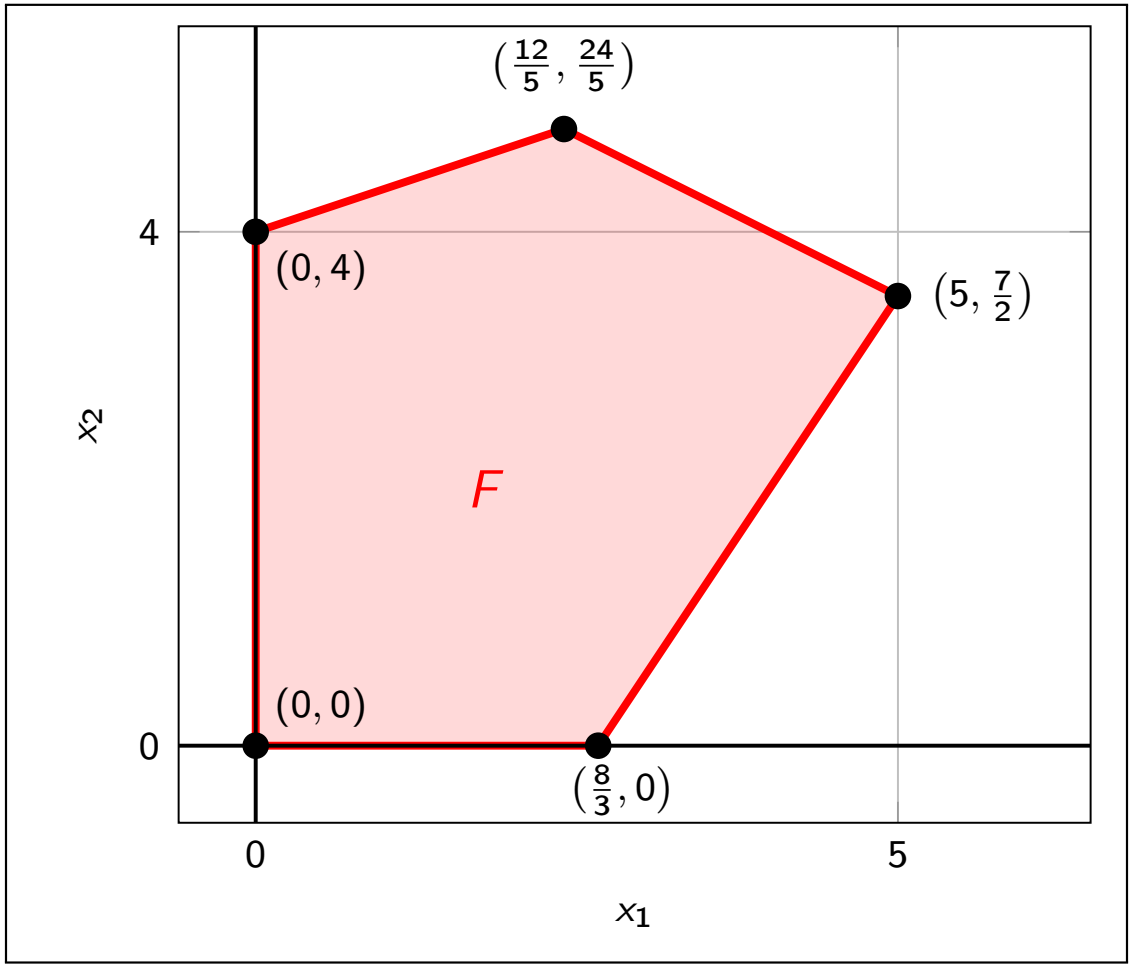
Example: faces, facets and vertices (1/3)

- Consider the feasible region F of the first LP problem of the dimension example
- It is full-dimensional, $\dim(F) = 2$, and then:
  1. its **facets** are its faces of dimension 1, that is, its five edges;
  2. its **vertices** are its faces of dimension 0, that is, its five corner points
- Each of the five linear constraints defines one facet, and then all of them are necessary to describe F

Example: faces, facets and vertices (2/3)

constraint	**facet it defines**	**vertices it contains**
$-x_1 + 3x_2 \leq 12$	edge from $(0, 4)$ to $\left(\frac{12}{5}, \frac{24}{5}\right)$	2
$x_1 + 2x_2 \leq 12$	edge from $\left(\frac{12}{5}, \frac{24}{5}\right)$ to $\left(5, \frac{7}{2}\right)$	2
$3x_1 - 2x_2 \leq 8$	edge from $\left(5, \frac{7}{2}\right)$ to $\left(\frac{8}{3}, 0\right)$	2
$x_2 \geq 0$	edge from $\left(\frac{8}{3}, 0\right)$ to $(0, 0)$	2
$x_1 \geq 0$	edge from $(0, 0)$ to $(0, 4)$	2

Example: faces, facets and vertices (3/3)



The five points marked in black are the vertices of F, and the five red segments joining them are its facets.

Summary: faces, facets and vertices

Faces:

- A face is the set of the points of F satisfying a valid inequality with equality
- Every constraint defines a face: the solutions for which it is binding

Facets and vertices:

- Facet: a face of dimension $\dim(F) - 1$, a constraint that cannot be removed
- Vertex: a face of dimension 0, a single point
- The vertices are exactly the **extreme points**: the points that do not lie strictly between two distinct points of F

Basic feasible solutions and vertices

Introduction: Basic feasible solutions and vertices

- We now relate the geometric notion of **vertex** to the algebraic notion of **basic feasible solution**, which is the one used by the simplex algorithm
- We consider an LP problem in standard maximization form with $n + m$ non-negative variables and m equality linear constraints, with $\text{rank}(\boldsymbol{A}) = m$
- We also see when several bases give the same vertex: **degeneracy**

Basic feasible solutions and vertices

Theorem 2: basic feasible solutions and vertices

Given an LP problem in standard maximization form and a point $\bar{\boldsymbol{x}} \in F$, the following two statements are equivalent:

1. $\bar{\boldsymbol{x}}$ *is a* ***basic feasible solution****, that is, there exists a basis* $\boldsymbol{A}_{\mathcal{B}}$ *such that* $\bar{\boldsymbol{x}}_{\mathcal{B}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{b} \geq 0$ *and* $\bar{\boldsymbol{x}}_{\mathcal{N}} = 0$*;*
2. $\bar{\boldsymbol{x}}$ *is a* ***vertex*** *of* F*.*

The idea of the proof

- The columns associated with the strictly positive entries of $\bar{\boldsymbol{x}}$ are linearly independent

if and only if

- $\bar{\boldsymbol{x}}$ cannot be moved in two opposite directions while staying in F

We use the theorem on vertices and extreme points:
a vertex is a point that is not the midpoint of two distinct points of F.

Proof (1/4)

- Let $P = \{ j \ : \ \bar{x}_j > 0 \}$
- $(1 \Rightarrow 2)$ Let $\bar{\boldsymbol{x}}$ be a basic feasible solution with basis $\mathcal{B}$
- Then $P \subseteq \mathcal{B}$, and then the columns $\{\boldsymbol{A}_j\}_{j \in P}$ are linearly independent, being a subset of the columns of the invertible matrix $\boldsymbol{A}_{\mathcal{B}}$
- Assume

$$\bar{\boldsymbol{x}} = \lambda \, \boldsymbol{x}^{(1)} + (1 - \lambda) \, \boldsymbol{x}^{(2)}$$

with $\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)} \in F$ and $\lambda \in (0, 1)$

Proof (2/4)

- For every $j \notin P$ we have $\bar{x}_j = 0$ and, since both points are non-negative, $x_j^{(1)} = x_j^{(2)} = 0$
- Setting $\boldsymbol{d} = \boldsymbol{x}^{(1)} - \boldsymbol{x}^{(2)}$, we obtain $d_j = 0$ for every $j \notin P$ and

$$\boldsymbol{A}\,\boldsymbol{d} = \boldsymbol{A}\boldsymbol{x}^{(1)} - \boldsymbol{A}\boldsymbol{x}^{(2)} = \boldsymbol{b} - \boldsymbol{b} = 0$$

$$\Longrightarrow \qquad \sum_{j \in P} d_j\, \boldsymbol{A}_j = 0$$

- By the linear independence of $\{\boldsymbol{A}_j\}_{j \in P}$ we get $\boldsymbol{d} = 0$, that is, $\boldsymbol{x}^{(1)} = \boldsymbol{x}^{(2)}$
- Accordingly $\bar{\boldsymbol{x}}$ is an extreme point and, by Theorem 1, a vertex of F

Proof (3/4)

- $(2 \Rightarrow 1)$ Let $\bar{\boldsymbol{x}}$ be a vertex of F and assume, by contradiction, that the columns $\{\boldsymbol{A}_j\}_{j \in P}$ are linearly dependent
- There exists $\boldsymbol{d} \neq 0$ with $d_j = 0$ for every $j \notin P$ and

$$\sum_{j \in P} d_j \boldsymbol{A}_j = 0$$

- Since $\bar{x}_j > 0$ for every $j \in P$, for $\varepsilon > 0$ small enough both points $\bar{\boldsymbol{x}} \pm \varepsilon \boldsymbol{d}$ are non-negative
- They satisfy $\boldsymbol{A}(\bar{\boldsymbol{x}} \pm \varepsilon \boldsymbol{d}) = \boldsymbol{b}$: they belong to F
- But $\bar{\boldsymbol{x}}$ is their midpoint, which contradicts Theorem 1

Proof (4/4)

- Accordingly $\{\boldsymbol{A}_j\}_{j \in P}$ are linearly independent
- Since $\text{rank}(\boldsymbol{A}) = m$, they can be completed to m linearly independent columns, whose indices form a basis $\mathcal{B} \supseteq P$
- With respect to it,

$$\bar{\boldsymbol{x}}_{\mathcal{N}} = 0 \qquad \text{and} \qquad \bar{\boldsymbol{x}}_{\mathcal{B}} = \boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{b} \geq 0$$

- $\bar{\boldsymbol{x}}$ is a basic feasible solution



Bases and vertices

- The correspondence between bases and vertices is **onto** but not necessarily **one-to-one**
- Every basic feasible solution is a vertex, but the same vertex may be associated with several different bases
- This happens only if the vertex is **degenerate**, that is, if some basic variable takes the value 0
- A non-degenerate vertex is associated with exactly one basis, formed by the indices of its strictly positive entries
- A degenerate vertex is usually, but not always, associated with several bases

Let us list all the bases of the first LP problem of the dimension example, and find its vertices.

Example: bases and vertices (1/4)

- Consider the first LP problem of the dimension example in standard maximization form, with $n + m = 5$ non-negative variables and $m = 3$ equality linear constraints
- A basis is a subset of 3 indices out of 5, and there are $\binom{5}{3} = 10$ such subsets
- All of them give an invertible matrix $\boldsymbol{A}_{\mathcal{B}}$, and then all of them are bases

Example: bases and vertices (2/4)

$\mathcal{B}$	$\boldsymbol{x}_{\mathcal{B}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{b}$	feasible	vertex (x_1, x_2)
$\{1,2,3\}$	$(5,\ \frac{7}{2},\ \frac{13}{2})$	✓	$(5, \frac{7}{2})$
$\{1,2,4\}$	$(\frac{48}{7},\ \frac{44}{7},\ -\frac{52}{7})$		
$\{1,2,5\}$	$(\frac{12}{5},\ \frac{24}{5},\ \frac{52}{5})$	✓	$(\frac{12}{5}, \frac{24}{5})$
$\{1,3,4\}$	$(\frac{8}{3},\ \frac{44}{3},\ \frac{28}{3})$	✓	$(\frac{8}{3}, 0)$
$\{1,3,5\}$	$(12,\ 24,\ -28)$		

Example: bases and vertices (3/4)

$\mathcal{B}$	$\boldsymbol{x}_{\mathcal{B}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\boldsymbol{b}$	feasible	vertex (x_1, x_2)
$\{1,4,5\}$	$(-12,\ 24,\ 44)$		
$\{2,3,4\}$	$(-4,\ 24,\ 20)$		
$\{2,3,5\}$	$(6,\ -6,\ 20)$		
$\{2,4,5\}$	$(4,\ 4,\ 16)$	✓	$(0,4)$
$\{3,4,5\}$	$(12,\ 12,\ 8)$	✓	$(0,0)$

Example: bases and vertices (4/4)

- Exactly 5 of the 10 bases are feasible, and they correspond to the 5 vertices of F determined in the previous example
- In this LP problem the correspondence is **one-to-one**, since no basic feasible solution is degenerate
- In every feasible row of the table all three basic variables are strictly positive

What happens instead at a degenerate vertex?

Let us look at a feasible region made of a single point.

Example: a degenerate vertex (1/3)

- Consider instead the following LP problem in canonical maximization form, whose feasible region consists of a single point:

$$
\begin{array}{rrrrrrrr}
\max & & x_1 & + & 2 & x_2 & & \\
\text{subject to} & & x_1 & + & & x_2 & \leq & 2 \\
& - & x_1 & - & & x_2 & \leq & -2 \\
& & x_1 & - & & x_2 & \leq & 0 \\
& - & x_1 & + & & x_2 & \leq & 0 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$

Example: a degenerate vertex (2/3)

- The first two constraints impose $x_1 + x_2 = 2$ and the last two impose $x_1 = x_2$, and then

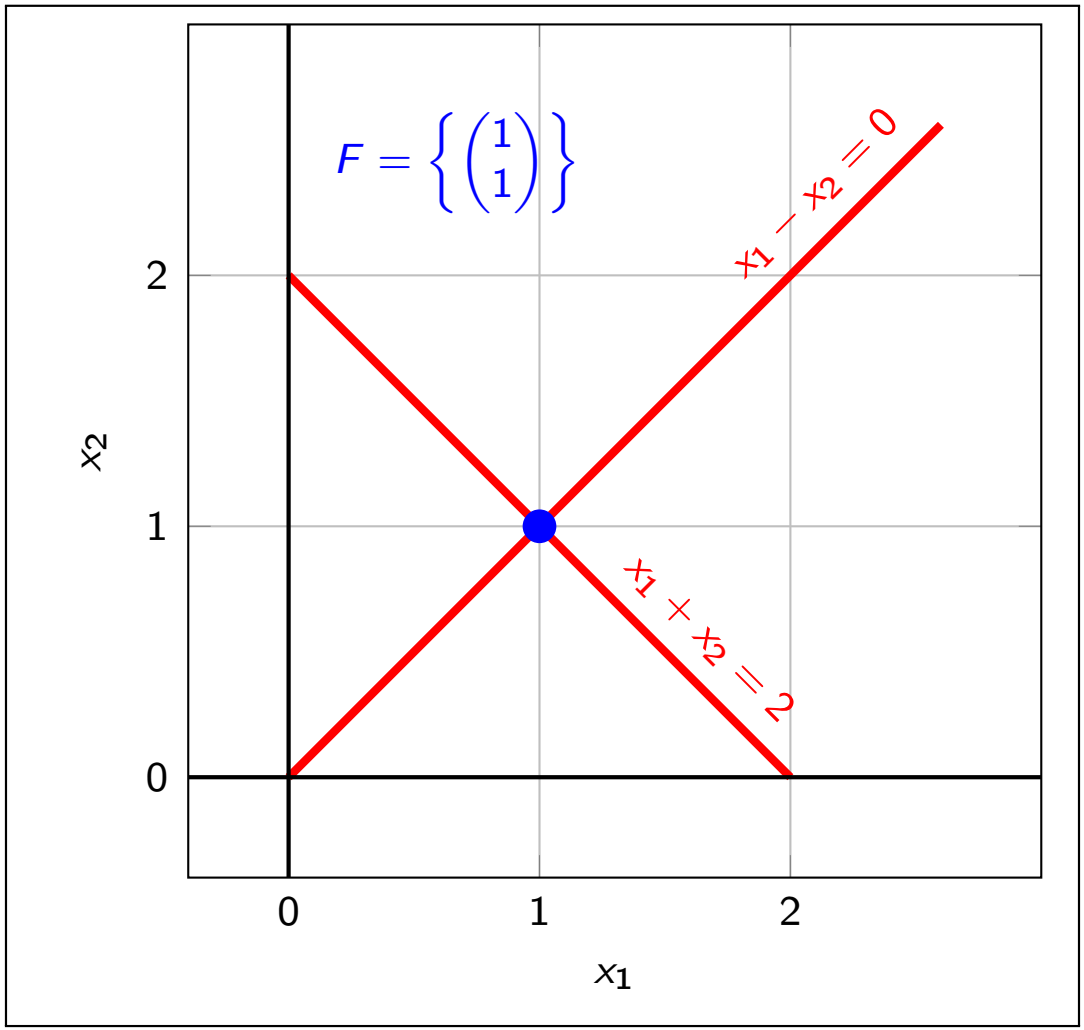
$$F = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}, \qquad \dim(F) = 0$$

- In standard maximization form the problem has 6 variables and 4 constraints, and the single vertex $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is associated with **four** different bases,

$$\{1, 2, 3, 5\}, \qquad \{1, 2, 3, 6\}, \qquad \{1, 2, 4, 5\}, \qquad \{1, 2, 4, 6\},$$

  in each of which two basic variables take the value 0
- This vertex is therefore **degenerate**, and the correspondence between bases and vertices is not one-to-one

Example: a degenerate vertex (3/3)



Summary: basic feasible solutions and vertices

Bases and vertices:

- A point of F is a basic feasible solution if and only if it is a vertex of F
- The columns of the strictly positive entries of a vertex are linearly independent

Degeneracy:

- Every basic feasible solution is a vertex, but a vertex may have several bases
- This happens only at a **degenerate** vertex, where some basic variable is 0
- A non-degenerate vertex has exactly one basis

The Fundamental Theorem, geometrically

Introduction: The Fundamental Theorem, geometrically

- The Fundamental Theorem of Linear Programming states that, whenever an LP problem admits an optimal solution, it admits a basic feasible solution which is optimal
- By Theorem 2, this statement has an immediate geometric reading
- We read the optimal solutions, the optimality check and the pivoting operation of the simplex algorithm on the polyhedron

Optimal vertices

Corollary 1: optimal vertices

*If an LP problem admits an optimal solution, then at least one **vertex** of its feasible region is an optimal solution.*

- This is the result that justifies the **graphical solution method**: to solve an LP problem with two variables it is enough to compute the objective value at each vertex of the feasible region and to keep the best one
- It is also the result that makes the simplex algorithm possible: the search for an optimal solution, which is a priori a search over infinitely many points, becomes a search over the finitely many vertices

The optimal face

- The set of the optimal solutions of an LP problem is itself a **face** of the feasible region
- It is the face defined by the valid inequality

$$\boldsymbol{c}'\boldsymbol{x} \leq z(\, LP\,)$$

- Accordingly:
  1. if that face is a vertex, the optimal solution is unique;
  2. if that face has dimension 1 or more, the LP problem admits infinitely many optimal solutions, and among them at least one vertex

The optimality check

- A vertex is an optimal solution if and only if no edge leaving it, bounded or unbounded, improves the objective function
- The **optimality check** of the simplex algorithm is based on this property: the non-basic variables correspond to the directions leaving the vertex, and their effect on the objective function is measured by the reduced costs
- The check is a sufficient condition for optimality but, at a degenerate vertex, not a necessary one
- A degenerate vertex may be optimal even if some reduced cost is positive, since the corresponding direction may immediately leave F

Adjacent vertices

Definition 7: adjacent vertices

Two vertices of a polyhedron are called ***adjacent*** *if the segment joining them is a face, that is, an edge.*

- In terms of bases, two non-degenerate basic feasible solutions are adjacent if and only if their bases differ by exactly one index

Pivoting and adjacent vertices

- A **pivoting operation** of the simplex algorithm replaces one index of the basis with another one
- If it is non-degenerate, it moves from one vertex to an **adjacent** one, along an edge of the feasible region, strictly increasing the objective value
- If it is degenerate ($t_m = 0$), it changes the basis but remains at the same vertex
- The simplex algorithm is, geometrically, a **walk on the edges** of the polyhedron

Example: the simplex algorithm on the edges (1/2)

- The execution of the simplex algorithm on the first LP problem of the dimension example, started from the basis $\mathcal{B} = \{3, 4, 5\}$, visits the following vertices:

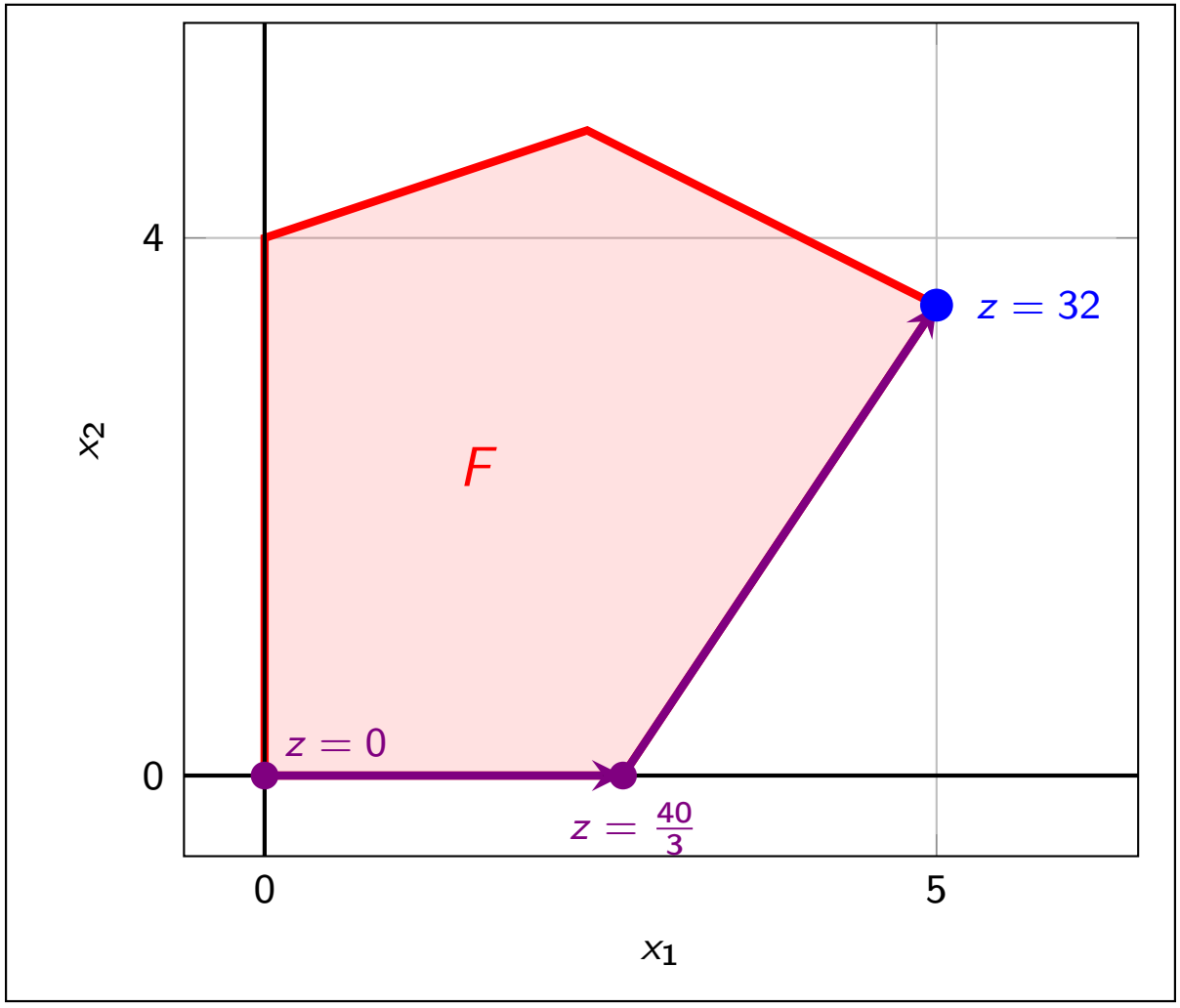
$$(0, 0) \longrightarrow \left(\frac{8}{3}, 0\right) \longrightarrow \left(5, \frac{7}{2}\right),$$

  with objective values 0, $\frac{40}{3}$ and 32

- Each step moves along one edge of F, to an adjacent vertex, and the objective value strictly increases

- In the figure on the next slide, in violet the two vertices visited before the last one, in blue the optimal vertex

Example: the simplex algorithm on the edges (2/2)



Summary: the Fundamental Theorem, geometrically

Optimal vertices:

- If an LP problem admits an optimal solution, at least one vertex is optimal
- The set of the optimal solutions is a face: the optimal face

The simplex algorithm on the polyhedron:

- Non-basic variables: directions leaving the vertex; reduced costs: their effect on the objective
- A non-degenerate pivoting operation moves to an **adjacent** vertex, along an edge
- The simplex algorithm is a walk on the edges of the polyhedron

Summary (1/3)

geometric notion	**algebraic counterpart**
polyhedron F	feasible region of the LP problem
$\dim(F)$	n minus the independent equalities valid on F
face	solutions making a valid inequality binding
facet	a constraint which cannot be removed
vertex = extreme point	basic feasible solution
degenerate vertex	several bases, one single point
edge between two vertices	pivoting operation
optimal face	set of the optimal solutions

Summary (2/3)

The number of vertices:

- The number of vertices of a polyhedron can be **exponential** in the number of its constraints and variables
- For an LP problem in standard maximization form it is bounded by $\binom{n+m}{m}$
- This is why the simplex algorithm, which visits one vertex at a time, cannot be shown to be efficient by simply counting the vertices
- And why its behaviour in practice is much better than its worst case

Summary (3/3)

- The **dimension** of a polyhedron is the number of its independent directions; in standard form $\dim(F) \leq n - \text{rank}(\boldsymbol{A})$
- **Faces** are defined by valid inequalities; **facets** have dimension $\dim(F) - 1$, **vertices** dimension 0
- Vertices = extreme points = **basic feasible solutions**; a degenerate vertex may have several bases
- If an optimal solution exists, an optimal **vertex** exists, and the simplex algorithm walks along the edges from vertex to adjacent vertex